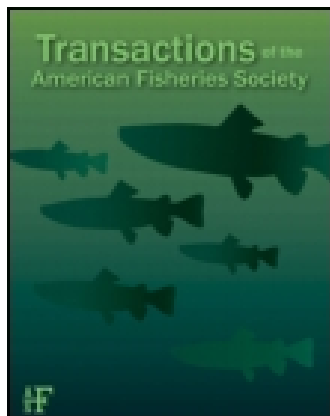




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Visual Prey Detection Responses of Piscivorous Trout and Salmon: Effects of Light, Turbidity, and Prey Size

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ARTICLE

Visual Prey Detection Responses of Piscivorous Trout and Salmon: Effects of Light, Turbidity, and Prey Size

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Abstract

Visual foraging models provide a useful framework for predicting distribution, foraging success, and predation risk in pelagic communities; however, the visual prey detection capabilities of different predator species within and among taxonomic groups have not been sufficiently evaluated. Our primary objective was to more adequately characterize variation in the reaction distances of piscivorous salmonids by evaluating important anadromous taxa. We measured reaction distances of yearling Chinook Salmon *Oncorhynchus tshawytscha* and adult Coastal Cutthroat Trout *O. clarkii clarkii* to fish prey over a range of prey sizes and ecologically relevant light and turbidity levels. Reaction distances of Coastal Cutthroat Trout increased rapidly with increasing light intensity (lx) and attained an average maximum of 187.1 cm above a light threshold of 18.0 lx. Reaction distances of Chinook Salmon increased at a slower rate to a maximum of 122.1 cm above a light threshold of 24.9 lx, declined exponentially with turbidity beyond a threshold of 1.65 NTU, and declined for prey sizes less than 50 mm FL. Reaction distances of Coastal Cutthroat Trout were consistently higher than those of Chinook Salmon across all light levels; this difference could not be attributed to the greater FLs of the Coastal Cutthroat Trout. Results from this and previous studies show that the functional form of reaction distance is similar across piscivorous salmonid species and life stages, but the magnitude of the response can vary considerably. Therefore, to adequately predict the strength of predation effects in pelagic communities, species- and life-stage-specific responses must be considered.

Dynamic environmental conditions can influence overlap between predators and prey in pelagic systems (Hardiman et al. 2004; Jensen et al. 2006). The manner in which perception and behavior interact with the physical environment to mediate piscivory after overlap is achieved is less well understood. Pelagic piscivores are primarily visual foragers (Loew and McFarland 1990). As a result, optical conditions can limit the foraging

success of piscivores, first by mediating their ability to detect prey (Beauchamp et al. 1999; Vogel and Beauchamp 1999) and second by affecting the capture rate of encountered prey (Petersen and Gadomski 1994; De Robertis et al. 2003; Mazur and Beauchamp 2003). By experimentally evaluating the influence of visibility on foraging behavior, we can identify mechanisms at scales that are pertinent to the perceptual fields of

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predators and prey and that help explain variability in the strength of predation effects. Here, we focus on visual prey detection and reaction by piscivorous salmonids.

Reaction distance—the average point at which a predator first exhibits a response to prey—is a valuable behavioral metric for measuring the effects of visibility on prey detection by piscivores (Howick and O'Brien 1983; Miner and Stein 1996; Vogel and Beauchamp 1999). Declining light and increasing turbidity reduce reaction distances, search volumes, and prey encounters at disproportionately higher rates for piscivores than for their planktivorous prey (Breck 1993; De Robertis et al. 2003). These asymmetric responses interact with dynamic light environments (Gjelland et al. 2009), basin productivity (Beauchamp et al. 1999), and other abiotic factors (e.g., temperature and oxygen; Hardiman et al. 2004; Hansen et al. 2013) to create complex patterns of predator–prey distributions, foraging success, and predation risk that affect the growth and survival of pelagic fishes over time and space (Jensen et al. 2006; Gjelland et al. 2009; Kahilainen et al. 2009). The influences of light intensity and scattering on reaction distance also appear to vary markedly among piscivorous species, suggesting that competition and intraguild predation within this functional group may be influenced by optical conditions (see Henderson and Northcote 1985; Mazur and Beauchamp 2003). However, these responses have been described for only a small number of species, thereby limiting the application and refinement of this approach.

Similarly, few studies have investigated the effect of prey fish size on the reaction distances of piscivores, and the available studies have generated potentially conflicting results. Largemouth Bass *Micropterus salmoides* exhibited increasing reaction distance with increasing sizes (30–80 mm FL) of prey (Bluegill *Lepomis macrochirus* and Redfin Shiner *Lythrurus umbratilis*; Howick and O'Brien 1983), whereas different sizes (55–139 mm FL) of salmonid prey did not affect the light-dependent reaction distances of Lake Trout *Salvelinus namaycush* (Vogel and Beauchamp 1999). For detection of prey, piscivores typically rely on the contrast between the prey and the background, whereas planktivorous fishes use an acuity-based system to detect small-bodied zooplankton (Breck 1993). Whether these differences among predator species result from the different visual capabilities or foraging modes ascribed to different taxonomic groups has yet to be determined. Nonetheless, there is presumably a size or a transitional range of prey sizes that affects prey fish detection and the reaction distance of piscivorous salmonids (Vogel and Beauchamp 1999).

Despite the importance of visibility in mediating predation rates in pelagic communities, the visual prey detection capabilities of anadromous salmonids are unknown. Chinook Salmon *Oncorhynchus tshawytscha* are important, visually oriented predators in predominately pelagic habitats of marine systems and some freshwater systems (e.g., Stewart and Ibarra 1991); they become piscivorous early in life and consume greater fractions of fish prey as they grow (Keeley and Grant 2001; Daly et al. 2009; Duffy et al. 2010). Recent evidence suggests that

foraging conditions promoting rapid growth of juvenile Chinook Salmon during early marine residence are critical for their overall marine survival (Duffy and Beauchamp 2011), and fish prey may play a key role in achieving such growth (Daly et al. 2009; Duffy et al. 2010). Therefore, it is particularly important to understand how visibility affects prey detection by early life stages of Chinook Salmon. Coastal Cutthroat Trout *O. clarkii clarkii* are top piscivores in many salmon-bearing waters (Trotter 1989; Cartwright et al. 1998; Nowak et al. 2004) and are important piscivores in nearshore coastal marine waters (Loch and Miller 1988; Duffy and Beauchamp 2008). We measured responses from these anadromous species to expand our understanding of the visual prey detection capabilities of piscivorous salmonids by complementing results from previous studies of resident freshwater taxa (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003).

Visual foraging models are useful tools for predicting distribution, foraging success, and predation risk in pelagic communities, either under ambient conditions or in response to future management or environmental alterations (Beauchamp et al. 1999). However, the utility of such models relies on sufficient parameterization of the prey detection process (Gerritsen and Strickler 1977; Aksnes and Giske 1993; Mazur and Beauchamp 2003). Our primary objectives were to more completely characterize variability in the visual prey detection capabilities of piscivorous salmonids by evaluating important anadromous forms and to address lingering uncertainties regarding the effects of prey size on reaction distance. In this study, we (1) experimentally measured the reaction distances of yearling Chinook Salmon and adult Coastal Cutthroat Trout over a range of ecologically relevant light and turbidity levels in a large laboratory tank; (2) parameterized and compared species-specific reaction distances as functions of light and turbidity; (3) evaluated whether the reaction distances of Chinook Salmon changed across the range of prey sizes that are consumed during early marine residence (Duffy et al. 2010) but that were smaller than the prey fish sizes examined in previous studies of piscivorous salmonids (Vogel and Beauchamp 1999); and (4) qualitatively compared our results with those from previous studies evaluating inland salmonids (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003).

METHODS

We measured the effects of light, turbidity, and prey size on reaction distances of piscivorous anadromous salmonids in freshwater tank experiments using video analysis. All experiments were conducted at the University of Washington's Big Beef Creek Field Station in Seabeck, Washington. We first measured the reaction distances of yearling Chinook Salmon (194–288 mm FL; $N = 20$) and Coastal Cutthroat Trout (255–440 mm FL; $N = 13$) across a range of light intensities (0.03–250 lx; measured at the water's surface) in clear water (turbidity < 0.5 NTU). The selected light levels resembled a

daylight–dusk–night cycle (Mazur and Beauchamp 2003). Next, we measured the reaction distances of Chinook Salmon (182–241 mm FL; $N = 36$) at incremental increases in turbidity (0.4–7.2 NTU) under a constant surface light level (50 lx). Trials were designed to detect the thresholds at which reaction distance began to decline and limit search volume (Mazur and Beauchamp 2003). Prey fish used for the light and turbidity experiments were Rainbow Trout *O. mykiss* (mean $\pm$ 2 SE = 49 ± 0.31 mm FL), as this species was used in previous studies (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003). Lastly, we measured reaction distances of Chinook Salmon (220–290 mm FL; $N = 12$) to a series of smaller prey fish sizes under low and high surface light levels (5 and 25 lx, respectively). Trials explored whether the contrast-based visual system of piscivores limited their reaction distances as prey size declined. We hypothesized that reaction distance would decline with prey size but that the decline would occur at a higher rate under low light levels given the likely greater difficulty in detecting smaller objects under more degraded visual conditions. Timing of prey availability and logistical constraints precluded the use of different sizes of a single prey species in these trials. Consequently, we used three species representing different prey sizes: Threespine Sticklebacks *Gasterosteus aculeatus* were used for the smallest prey size treatment (23 ± 0.77 mm FL), juvenile Coastal Cutthroat Trout were used for the intermediate prey size treatment (35 ± 0.62 mm FL), and Rainbow Trout were used for the largest prey size treatment (50 ± 0.59 mm FL).

Collection, maintenance, and acclimation of experimental fish.—Yearling Chinook Salmon were obtained in spring 2010 and 2011 from Hoodsport Hatchery (Washington Department of Fish and Wildlife [WDFW]), Hoodsport, Washington. Wild Coastal Cutthroat Trout were captured via hook and line from Big Beef Creek near its mouth on Hood Canal, Washington, during summer through fall in 2009–2011. The captured Coastal Cutthroat Trout were held in outdoor circular tanks (4-m diameter) covered with netting and black-mesh cloth to reduce the influence of direct sunlight. These tanks were supplied with thermally stable (9.5–11.5°C) well water and were subjected to a natural photoperiod throughout the duration of experimentation. Chinook Salmon were held indoors under identical conditions except that incandescent lights on a timer were used to mimic the natural photoperiod. Chinook Salmon were maintained on formulated feed (Silver Cup Fish Feed). Coastal Cutthroat Trout were maintained on a combination of formulated feed and frozen krill (Hikari Bio-Pure). We conditioned the predators to feed on live prey fish for 1–5 months prior to experimentation. Rainbow Trout prey were obtained from Eells Springs Hatchery (WDFW), Shelton, Washington; Coastal Cutthroat Trout prey were spawned from the adults captured in Big Beef Creek and were raised in the hatchery; and wild Threespine Stickleback prey were captured from the settling ponds at the Big Beef Creek Field Station. Animal care and handling for this research were performed under the auspices of the University of Washington's Institutional Animal Care and Use Committee Protocol Number 3286-19.

Experimental arena.—Reaction distances were measured in an indoor circular tank (4.1-m diameter $\times$ 1.2 m high) that was filled with water to a depth of 0.5 m. The diameter of our circular arena exceeded (by 4–8 times) the maximum average reaction distances ($\sim$ 0.5–1.0 m) measured for piscivorous salmonids in previous studies (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003). Additionally, unlike the narrow rectangular tanks used previously (4.5 m long $\times$ 1.0 m wide $\times$ 1.0 m high), our arena provided a greater breadth of angles from which the predators could orient toward the prey fish. The arena was lined with a flexible gray polyvinyl chloride material as was done in previous experiments (Vogel and Beauchamp 1999). A rectangular curtain made from the same material was used to split the arena into halves while predators were acclimating. Prey were tethered inside one of four acrylic tubes (14-cm diameter) to eliminate nonvisual stimuli. Two tubes were positioned vertically near the tank wall and about 1.0 m apart at the far end of the arena on both sides of the opaque acclimation curtain. The tether consisted of low-visibility fishing line (4.5-kg test; 0.20-mm diameter) that extended through each tube and was held taut by a small weight.

The arena was illuminated by six fluorescent fixtures (two lamps per fixture) that were suspended 2.4 m above the surface of the water. We chose lamps (F32-T8-TL865 PLUS ALTO; Philips Lighting) that were designed to mimic natural daylight but with a higher color temperature (6,500 kelvins) so that the spectral composition of emitted light was dominated by violet, blue, and green (range = 380–760 nm; Figure 1). These lamps best represented the underwater light environment in the types of systems inhabited by salmonids (Horne and Goldman 1994; Kirk 2011) and matched the rapid drop in sensitivity of fishes to longer (red) wavelengths (Horodysky et al. 2010; Figure 1). Light intensity was controlled by adding multiple layers of fiberglass window screen (Vogel and Beauchamp 1999) between the lamps and a diffuser plate on each fixture and then was fine-tuned by using a series of dimming switches (Lutron Electronics, Inc.). Turbidity was controlled by mixing consistent amounts of pulverized kaolin clay (Acros Organics; 50–62% of particles were $< 2 \mu\text{m}$) into the arena with a submersible water pump. The arena was shrouded with a layer of black plastic sheeting to exclude light from external sources.

Trials were recorded by using two fixed-focus, low-light, black-and-white security cameras (Model CVC-321WP; Speco Technologies) that were mounted perpendicular to and 2.6 m above the surface of the water. Cameras were connected to a video capture card (Model PV-183-8; Bluecherry) in a desktop computer. Video acquisition software was from the Nonlinear Dynamics and Control Laboratory at the University of Washington; this software synchronized each camera and recorded video at a rate of 60 frames/s with a resolution of 640×480 pixels. To enhance camera sensitivity for trials at low light levels (≤ 1 lx), we recorded video with the addition of infrared light (850 nm, not detectable by humans or fish; Douglas and Hawryshyn 1990; Mazur and Beauchamp 2003) from six strategically placed illuminators (Model CM126-30; Scene Electronics Co., Ltd.).

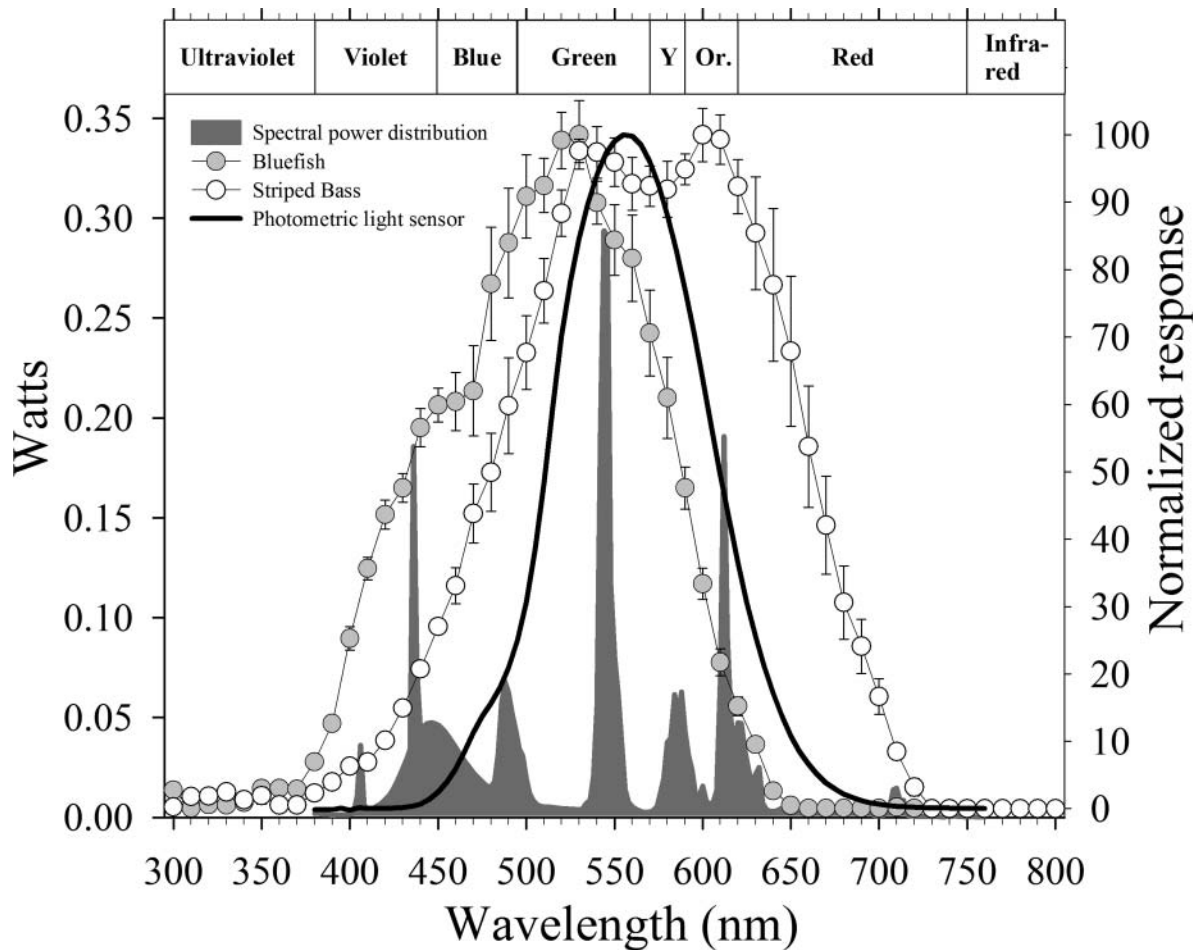


FIGURE 1. Spectral power distribution (W) of the fluorescent lamps at maximum voltage (data from Philips Lighting) in relation to the normalized spectral sensitivity curves developed for Bluefish *Pomatomus saltatrix* and Striped Bass *Morone saxatilis* (circles are means of five individuals $\pm$ 1 SE; data from Horodysky et al. 2010) and the normalized spectral responsivity curve of the LI-210 photometric light sensor (measures lx; data from LI-COR; Y = yellow, Or. = orange).

Experimental protocol and measurement of treatment variables.—The predators were tested in pairs; to enhance their motivation to feed, the predators were deprived of food for at least 36 h prior to use in an experimental trial (Meager et al. 2005). Placement of multiple predators in the arena improved the fish's willingness to explore the arena (De Robertis et al. 2003). All predators were allowed to hunt free-swimming prey fish in the arena before being used in trials. Experience levels were controlled by cycling the predators through a series of "used" and "unused" holding tanks. Based on variability observed among trials in previous experiments (Vogel and Beauchamp 1999), two to four sets of predators were tested at each surface light level (0.03, 0.10, 1, 5, 10, 15, 20, 25, 50, and 250 lx; $N = 41$ for Chinook Salmon, $N = 31$ for Coastal Cutthroat Trout) and each turbidity level (near 0.5, 1.0, 1.5, 2, 3, 4, 5, 6, and 7 NTU; Chinook Salmon only: $N = 30$). Six pairs of Chinook Salmon were tested at each combination of light level and prey size ($N = 36$). Treatments were blocked such that one trial of each treatment

or combination thereof was completed in random order before replication as designated by the blocks to reduce time-varying effects.

For each trial, a single, live prey fish was tethered through the connective tissue underneath the maxilla (large prey) or through the lower jaw (small prey) and was placed inside one of four acrylic tubes in the middle of the water column. The tube that contained the prey fish was randomly selected for each trial. The tethering procedure allowed the prey to ventilate and rotate freely around a central pivot point (advantageous because the movement of prey often elicits responses from predators; Howick and O'Brien 1983) while maintaining a fixed position to standardize measurements of reaction distance. Preliminary experiments (using 50-mm prey in clear water at 10 and 50 lx) showed no significant differences between reaction distances measured in relation to prey that were tethered outside of the tubes versus inside the tubes ($F_{3,37} = 1.25$, $P = 0.271$). Predators were placed on the opposite side of the opaque curtain from

the prey and were allowed to acclimate to the light conditions for 1 h to ensure light–dark adaptation (Ali 1959). During the turbidity trials, we reduced the acclimation period to 30 min to minimize the amount of kaolin that settled out of suspension. After acclimation, the curtain was lifted and the predators were allowed to respond to the tethered prey for 1 h.

Before and after each trial, light levels were measured at the water's surface from six locations around the perimeter of the arena by using a calibrated LI-210 photometric sensor (cosine corrected; LI-COR) and an LI-1400 data logger. We report the average of pretrial and post-trial measurements ($N = 12$) to account for any minor deviations from the target irradiance (mean change ± 2 SE = $4.1 \pm 0.76\%$) that occurred during a trial. The photometric sensor measured visible radiation in lux (380–770 nm) using the spectral responsivity of an average human eye (Figure 1). We also recorded corresponding measurements of photosynthetically active radiation (400–700 nm; microeinsteins [$\mu\text{E}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$] by using an LI-190 terrestrial quantum sensor (cosine corrected; LI-COR). Unlike the photometric sensor, the quantum sensor responded equally to all photons across the 400–700-nm range. The relationship ([visible radiation, lx] = $66.849 \times$ [photosynthetically active radiation, $\mu\text{E}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$]) between units under our fluorescent lamps was strongly linear ($N = 1,350$; $r^2 = 0.99$). The bottom of the arena reflected light back up into the water column, causing irradiance underwater to be slightly higher than that indicated by the surface light measurements. Therefore, a linear model (subsurface light = $1.206 \times$ surface light; $N = 169$; $r^2 = 0.99$), which was generated by pairing the surface light measurements from the LI-190 quantum sensor with subsurface light measurements from an LI-193 underwater spherical quantum sensor, was used to correct all mean surface measurements to the light conditions experienced by predators at the depth of the prey.

Turbidity (NTU) was measured with a LaMotte Model 2020e turbidity meter. Before and after each trial, a single water sample was integrated from three mid-water-column points around the perimeter of the arena. We measured 5–8 subsamples of the pretrial and post-trial water samples; the means from the pretrial and post-trial measurements were averaged to account for clay settling out of suspension over the duration of each trial. Clay settled by an average (± 2 SE) of $30 \pm 6\%$ over the 1.5-h-long combined acclimation and trial period. Turbidity levels were highly reproducible ([turbidity, NTU] = $0.495 \times$ [kaolin concentration, g/m^3] + 0.51 ; $r^2 = 0.99$).

Oceanographic studies generally quantify turbidity in terms of beam attenuation (the sum of absorption and scatter; Kirk 2011). Therefore, we developed a conversion from turbidity in NTU to beam attenuation for our kaolin suspensions. The percentage of light at a wavelength of 660 nm transmitted through a 10-mm cuvette was measured by using a Spectronic 21 DV spectrophotometer (Milton Roy). Beam attenuation was calculated by using the standard formula $T = e^{-cr}$, where T (%) is the light transmitted through path length r (m) at attenuation rate c (m^{-1} ; Kirk 2011). Beam attenuation of the kaolin suspensions

was highly correlated with turbidity across a range of 0–10 NTU ($c = 0.40 \times$ [turbidity, NTU]; $N = 26$; $r^2 = 0.98$).

Camera model and estimation of reaction distance.—Reaction distances were estimated from video recordings by using two-dimensional camera tracking techniques. By using a 0.5-m water depth, we assumed that the predators always reacted to the tethered prey fish on the same two-dimensional plane in the middle of the water column, thus simplifying measurements. This assumption was supported by the approximate position of the predators in the water column relative to the prey, as determined based on the fish's shadows that were created on the bottom of the arena (Laurel et al. 2005).

Nodes of a 20×20 -cm reference grid ($N = 324$) were marked on the bottom of the experimental arena to calibrate the overhead cameras. Using ImageJ version 1.45s, we linked the coordinates of each node in pixels taken from still images with the known coordinates of each node in centimeters based on a defined grid origin. With this information, we fitted a series of third-order polynomial regression models (as applied by Hughes and Kelly 1996) that converted the coordinates in pixels (x_c, y_c) taken from anywhere within the image of the reference grid to the known coordinates in centimeters (x_g, y_g). The set of regression models that were fitted to each camera specifically included

$$x_g = p_1 + p_2x_c + p_3y_c + p_4x_c^2 + p_5x_cy_c + p_6y_c^2 + p_7x_c^3 + p_8x_c^2y_c + p_9x_cy_c^2 + p_{10}y_c^3 \quad (1)$$

and

$$y_g = p_1 + p_2x_c + p_3y_c + p_4x_c^2 + p_5x_cy_c + p_6y_c^2 + p_7x_c^3 + p_8x_c^2y_c + p_9x_cy_c^2 + p_{10}y_c^3, \quad (2)$$

where p_1 – p_{10} were fitted parameters. Parameters were estimated by using least squares (unadjusted r^2 for all fits of the model > 0.998 ; computed as $1 - [\text{SS}_{\text{residual}}/\text{SS}_{\text{total}}]$, where SS = sum of squares) with a multidimensional optimization routine (optim function) in R version 2.14.1 (R Development Core Team 2011).

We viewed the video recordings with Media Player Classic Home Cinema version 1.5.1.2903 (freeware). When a reaction was noted, the coordinates in pixels (x_c, y_c) from the predator's position (eye region of the head) in the associated video frame were recorded using ImageJ; the coordinates in pixels were then converted to coordinates in centimeters (x_g, y_g) by using the regression models fitted to the camera that captured the reaction. Because the position of the tethered prey was known, we simply used the distance formula to estimate the reaction distance. The potential error in any reaction distance measurement that was obtained using this technique was largely within $\pm 5\%$ (98.5% of the observations), as characterized by the distribution of percentage deviations between the known and predicted distances from all grid nodes to each tether location (pooled across

each camera; $N = 1,772$; mean absolute percentage ± 2 SE = $0.92 \pm 0.10\%$).

Statistical analyses.—We used a model selection approach to analyze the reaction distances of Chinook Salmon and Coastal Cutthroat Trout as a function of light, turbidity, and prey size. A candidate set of plausible models was fitted to the mean reaction distances that were calculated from each trial (responses pooled across predators; Vogel and Beauchamp 1999; Mazur and Beauchamp 2003). For light, we selected four biologically meaningful functional relationships that could capture the plateau response observed in reaction distance (see Results).

We included the two primary functions used in previous studies (Mazur and Beauchamp 2003): (1) a linear hockey stick including a breakpoint and (2) a similar two-piece model in which the increasing limb was a nonlinear power function. We also tested two continuous functions that exhibited asymptotic behavior: (1) a power function and (2) a Holling type II functional response (Holling 1959) that included a y-intercept to improve its comparability with the other models (Table 1). When analyzing the Chinook Salmon turbidity experiments, we only considered a declining exponential function (Miner and Stein 1996; Vogel and Beauchamp 1999). We tested for a significant

TABLE 1. Results from the model selection analysis evaluating different functional relationships (model and $f(x)$ columns; I = light intensity [lx]) for describing light-dependent reaction distances (RD) of piscivorous Chinook Salmon and Coastal Cutthroat Trout. For each model formulation tested, the corresponding values of the difference in Akaike's information criterion corrected for small sample size (ΔAIC_c) and Akaike weight (w_i) are presented. These values are referenced to the results generated from different formulations within each candidate model (within-model columns) and across the entire set of candidate models (across-models columns). Results within each candidate model are listed beginning with the most saturated formulation (i.e., in which no parameters are shared between species; “—” under the shared parameters column) and ending with the most reduced formulation (i.e., in which all parameters are shared between species). Model parameters are represented by a , b , and c ; l = log-likelihood. The total number of parameters (k) corresponding to each formulation includes the species-specific error terms. Values in bold italics correspond to the best-fitting models (i.e., ΔAIC_c values ≤ 2).

Model	$f(x)$	Shared parameters	k	$-2l$	AIC_c	ΔAIC_c		w_i	
						Within model	Across models	Within model	Across models
Hockey stick (linear)	RD = $a + bI$ for $I \leq t$	—	8	574.377	592.663	0.00	0.00	0.9175	0.3669
	RD = $a + bI$ for $I > t$	a	7	588.133	603.883	11.22	11.22	0.0034	0.0013
		b	7	593.429	609.179	16.52	16.52	0.0002	0.0001
		t	7	581.841	597.591	4.93	4.93	0.0780	0.0312
		a, b	6	626.317	639.610	46.95	46.95	0.0000	0.0000
		a, t	6	601.233	614.526	21.86	21.86	0.0000	0.0000
		b, t	6	593.245	606.537	13.87	13.87	0.0009	0.0004
		a, b, t	4	706.009	714.606	121.94	121.94	0.0000	0.0000
Piecewise (power)	RD = aI^b for $I \leq t$	—	8	580.453	598.738	4.11	6.08	0.0671	0.0176
	RD = aI^b for $I > t$	a	7	613.444	629.194	34.57	36.53	0.0000	0.0000
		b	7	580.452	596.202	1.57	3.54	0.2386	0.0625
		t	7	581.129	596.879	2.25	4.22	0.1701	0.0446
		a, b	6	652.945	666.238	71.61	73.57	0.0000	0.0000
		a, t	6	615.929	629.221	34.59	36.56	0.0000	0.0000
		b, t	6	581.336	594.628	0.00	1.97	0.5241	0.1373
		a, b, t	4	706.608	715.205	120.58	122.54	0.0000	0.0000
Power	RD = aI^b	—	6	614.529	627.822	2.38	35.16	0.2331	0.0000
		a	5	648.964	659.873	34.43	67.21	0.0000	0.0000
		b	5	614.530	625.439	0.00	32.78	0.7669	0.0000
		a, b	3	716.125	722.478	97.04	129.82	0.0000	0.0000
Holling type II (with y-intercept)	RD = $aI/(b + I) + c$	—	8	576.145	594.431	0.26	1.77	0.4485	0.1516
		a	7	589.590	605.340	11.17	12.68	0.0019	0.0006
		b	7	578.425	594.175	0.00	1.51	0.5097	0.1723
		c	7	583.603	599.353	5.18	6.69	0.0383	0.0129
		a, b	6	593.796	607.088	12.91	14.42	0.0008	0.0003
		a, c	6	625.682	638.974	44.80	46.31	0.0000	0.0000
		b, c	6	593.859	607.151	12.98	14.49	0.0008	0.0003
		a, b, c	4	706.762	715.359	121.18	122.70	0.0000	0.0000

turbidity threshold by fitting the model with and without a breakpoint; in the model that included a breakpoint, the reaction distance was a constant prior to the breakpoint and declined exponentially thereafter.

For light level, different formulations of each candidate model were generated by iteratively sharing parameters between Chinook Salmon and Coastal Cutthroat Trout, starting with the most saturated formulation (i.e., a fully parameterized function fitted to the data for each species) and ending with the most reduced formulation (i.e., a single function fitted to the data combined across species). Models were fitted to the data by using maximum likelihood estimation in R (mle2 function within the bbmle package; Bolker 2012). Error terms were assumed to be normally distributed and were estimated separately for each species in all model formulations. We used the same methods to fit the turbidity model to the data for Chinook Salmon. This procedure allowed us to test for significant differences in parameter estimates between the species and to test for a significant turbidity threshold directly via model selection. We used Akaike's information criterion corrected for small sample size (AIC_c ; Burnham and Anderson 2002) to select the best model.

The length distributions of Chinook Salmon and Coastal Cutthroat Trout predators were overlapping but not identical, thus limiting our ability to determine whether observed differences in reaction distance resulted from a species effect or an ontogenetic effect based on the model selection results alone. Therefore, we used linear models (lm function in R) with species as a grouping factor and FL as a continuous explanatory variable to test whether mean reaction distance under nonlimiting light conditions differed between the two species after taking predator length into account. Reaction distances were pooled and FLs were averaged for each Chinook Salmon pair, but reaction distances and FLs were analyzed separately for each Coastal Cutthroat Trout because these fish were individually identifiable in the video recordings. Reaction distance data were centered (on FL) and standardized (z -transformed) prior to fitting the linear models (Schielzeth 2010). We started with a linear model that contained all effects (including the interaction term), and then we iteratively reduced the model terms based on AIC_c (Burnham and Anderson 2002).

Data from the prey size experiment were analyzed via the same model selection procedures used in the predator length analysis. We used linear models to test whether reaction distances of Chinook Salmon varied as a function of prey size (continuous explanatory variable) and light (coded as a factor level) and whether the effect of prey size depended on light level (interaction term). Models were fitted to the mean reaction distances ($N = 6$) measured at each combination of prey size and light level. Both reaction distance and prey size (FL) were $\log_{10}$ transformed to improve linearity. We tested for a block effect by using simple linear regression. Reaction distances showed no temporal trend during the course of the experiment at an α of 0.05 (i.e., no significant block effect; slope = -4.08 , $r^2 = 0.076$, $P = 0.10$).

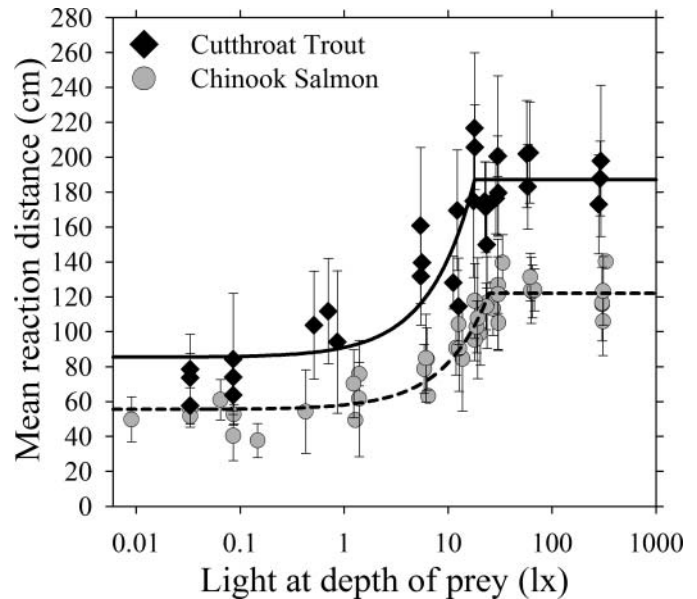


FIGURE 2. Reaction distance as a function of light for piscivorous Chinook Salmon ($N = 41$) and Coastal Cutthroat Trout ($N = 31$) that were presented with Rainbow Trout prey (47–52 mm FL). Data points represent the mean reaction distance (± 2 SE) pooled across predators from each individual trial. Lines represent the fitted piecewise models (solid line for Coastal Cutthroat Trout; dashed line for Chinook Salmon). Note that the x -axis (light) is presented on a $\log_{10}$ scale.

RESULTS

Reaction Distance as a Function of Light

Overall, reaction distances of Coastal Cutthroat Trout were higher than those of Chinook Salmon across all light levels (Figure 2). The best-fitting model formulation ($\Delta AIC_c = 0$) for describing reaction distance as a function of light was the fully saturated (i.e., no parameters shared between species), piecewise linear hockey stick (Table 1). Based on the fitted equations, reaction distance was strongly dependent on light for both Chinook Salmon ($r^2 = 0.89$) and Coastal Cutthroat Trout ($r^2 = 0.84$; Table 2; Figure 2). For both species, reaction distance increased rapidly with light to a breakpoint termed the saturation intensity threshold (SIT; Henderson and Northcote 1985), and these breakpoints differed significantly between the species (24.9 lx for Chinook Salmon; 18.0 lx for Coastal Cutthroat Trout; Table 2) based on the model selection results (Table 1). As light increased from low levels to the SIT, the reaction distance of Coastal Cutthroat Trout increased at a significantly higher rate (slope = 5.66) and reached a much higher maximum (187.1 cm) than the reaction distance of Chinook Salmon (slope = 2.68; maximum = 122.1 cm; Table 2). After controlling for predator FL, significant differences in the reaction distance above the SIT (to remove the effect of light) were still apparent between the species (Figure 3). The best-fitting linear model included species-specific y -intercept terms but a shared slope equal to zero, thus supporting a strong species difference with no effect

TABLE 2. Best-fitting models developed for the reaction distances (RD; cm) of piscivorous Chinook Salmon and Coastal Cutthroat Trout as functions of light intensity (I ; lx), turbidity (NTU), and prey size (FL). An equation describing the proportional decline in RD ($P[RD_{\max}]$, where $RD_{\max}$ is maximum RD) with turbidity is also shown for Chinook Salmon. The models fitted to the RD of Chinook Salmon as a function of prey size and light (SIT = saturation intensity threshold) in log-log space are presented in linear form.

Variable	Limb or factor level	Model	Breakpoint	r^2	Parameter error	
					Parameter	SE
Chinook Salmon						
Light	Increasing Level	$RD = 2.68I + 55.39$	≤ 24.9 lx	0.89	Slope	0.222
		$RD = RD_{\max} = 122.11$	> 24.9 lx		y-intercept	2.608
					Breakpoint	1.819
Turbidity	Level Declining	$RD = RD_{\max} = 124.06$	≤ 1.65 NTU	0.96	y-intercept	2.426
		$RD = 184.40e^{(-0.240 \cdot NTU)}$	> 1.65 NTU		Exponent	0.015
					Breakpoint	0.146
	Level Declining	$P(RD_{\max}) = 1.0$	≤ 1.65 NTU	0.96		
		$P(RD_{\max}) = 1.49e^{(-0.240 \cdot NTU)}$	> 1.65 NTU			
Prey size	Low light (<SIT)	$RD = (FL^{0.415})(10^{1.288})$		0.65	Exponent 1 (low)	0.124
	High light (>SIT)	$RD = (FL^{0.709})(10^{0.935})$			Exponent 2 (low)	0.191
					Exponent 1 (high)	0.124
					Exponent 2 (high)	0.190
Coastal Cutthroat Trout						
Light	Increasing Level	$RD = 5.66I + 85.31$	≤ 18.0 lx	0.84	Slope	0.477
		$RD = RD_{\max} = 187.05$	> 18.0 lx		y-intercept	6.105
					Breakpoint	0.543

of FL and no interaction between species and FL (Table 3). Reaction distances of Coastal Cutthroat Trout at higher light levels were 46% greater than those of Chinook Salmon.

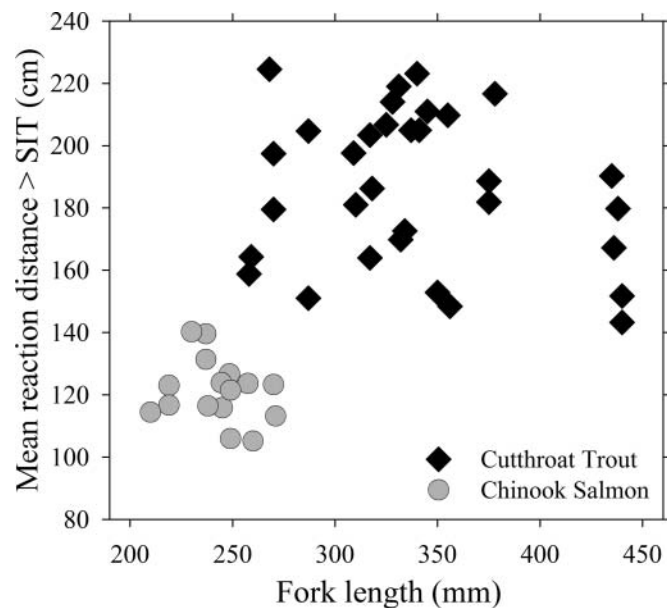


FIGURE 3. Mean reaction distances (cm) as a function of predator FL for pairs of Chinook Salmon ($N = 16$) and individual Coastal Cutthroat Trout ($N = 32$) that were presented with Rainbow Trout prey (47–52 mm FL) at light levels above the saturation intensity threshold (SIT; lx).

Reaction Distance as a Function of Turbidity

Reaction distances of Chinook Salmon were strongly dependent on turbidity ($r^2 = 0.96$; Table 2). The best-fitting model for describing reaction distance as a function of turbidity included a breakpoint (1.65 NTU) beyond which reaction distance began to decline exponentially (Table 2; Figure 4). In contrast, the model that excluded the breakpoint fit poorly ($\Delta AIC_c = 22.65$). Reaction distance decreased by about 70% from the breakpoint to 7.2 NTU.

Reaction Distance and Prey Size Effects

The best-fitting linear model for describing reaction distance as a function of prey size and light included all terms, suggesting that (1) light was an important factor, (2) reaction distance varied as a function of prey size, and (3) the extent to which reaction distance varied as a function of prey size depended on light level (Table 4). Based on the model selection results, reaction distance declined significantly with decreasing prey size at both the low and high light levels, but the rate of this decline was significantly greater at the high light level (slope = 0.71) than at the low light level (slope = 0.42). The significant interaction term was driven by the convergence of reaction distances measured at low and high light levels for the smallest prey size examined (~23-mm Threespine Sticklebacks; Figure 5).

DISCUSSION

Our experiments show that the functional form of reaction distance over ecologically relevant levels of light and turbidity

TABLE 3. Results from fitting different linear models to test whether reaction distances (RD) above the saturation intensity threshold differed between piscivorous Chinook Salmon and Coastal Cutthroat Trout after accounting for predator FL. Values in bold italics (ΔAIC_c = difference in Akaike's information criterion corrected for small sample size; w_i = Akaike weight) represent the best-fitting models (i.e., ΔAIC_c values ≤ 2). The total number of parameters (k) in each linear model includes the error term (ϵ); l = log-likelihood.

Linear model	k	$-2l$	AIC_c	ΔAIC_c	w_i
RD ~ Species + ϵ	3	78.016	84.561	0.00	0.4841
RD ~ Length + Species + ϵ	4	76.011	84.941	0.38	0.4003
RD ~ Length + Species + (Length $\times$ Species) + ϵ	5	75.998	87.427	2.87	0.1155
RD ~ Length + ϵ	3	134.607	141.153	56.59	0.0000

is similar across species and life stages of piscivorous salmonids that inhabit both marine and freshwater systems; however, the magnitude of the responses can differ. The reaction distance of Coastal Cutthroat Trout increased rapidly with increasing light level to a maximum beginning at 18.0 lx. The reaction distance of yearling Chinook Salmon increased at a slower rate to a maximum beginning at 24.9 lx, declined exponentially with increasing turbidity beyond a low turbidity threshold (1.65 NTU) when light levels were above the SIT, and decreased with decreasing prey size. Previous experiments with adult piscivorous Lake Trout, Rainbow Trout, and Bear Lake-strain Bonneville Cutthroat Trout *O. clarkii utah* resulted in the same plateau response and similar SIT values ranging from 17.00 to 18.75 lx (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003; Figure 6). Results from those studies also suggested that the threshold value beyond which turbidity limits reaction distance

ranged from 1 to 2 NTU, but this response has not been measured precisely until now. Lastly, reaction distances of Lake Trout in the prior studies did not change with prey size over the 55–139-mm length range (Vogel and Beauchamp 1999), but the investigators did not evaluate whether this relationship held for the smaller prey sizes that are more relevant for juvenile life stages. We found that the reaction distance of yearling Chinook Salmon was significantly reduced for prey smaller than 50 mm. Therefore, models of encounter rate for juvenile piscivores may benefit from the inclusion of prey size data.

The linear hockey stick model provided the best fit to the light-dependent reaction distances for both Chinook Salmon and Coastal Cutthroat Trout. However, the other piecewise model (which utilized a power function for the increasing limb) and two formulations of the modified Holling type II functional response also fit the data well ($\Delta AIC_c \leq 2$; Burnham and Anderson 2002).

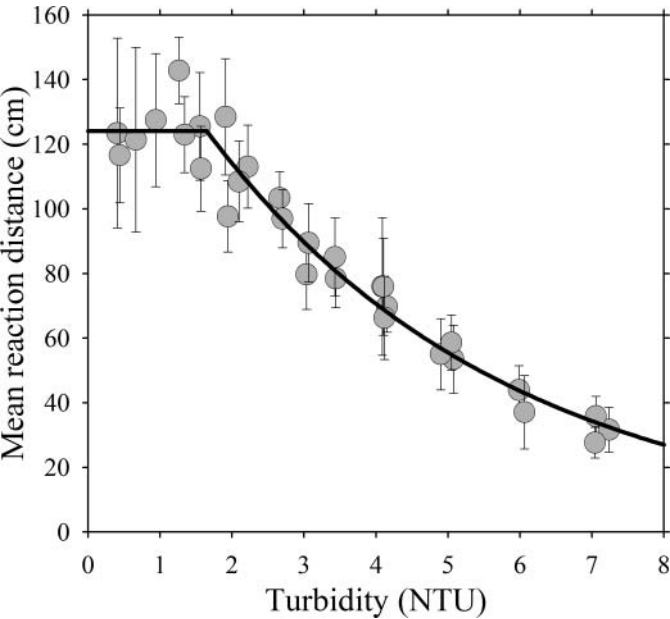


FIGURE 4. Reaction distance as a function of turbidity for piscivorous Chinook Salmon ($N = 30$) that were presented with Rainbow Trout prey (48–54 mm FL). Data points represent the mean reaction distance (± 2 SE) pooled across predators from each individual trial. The solid line represents the fitted piecewise model.

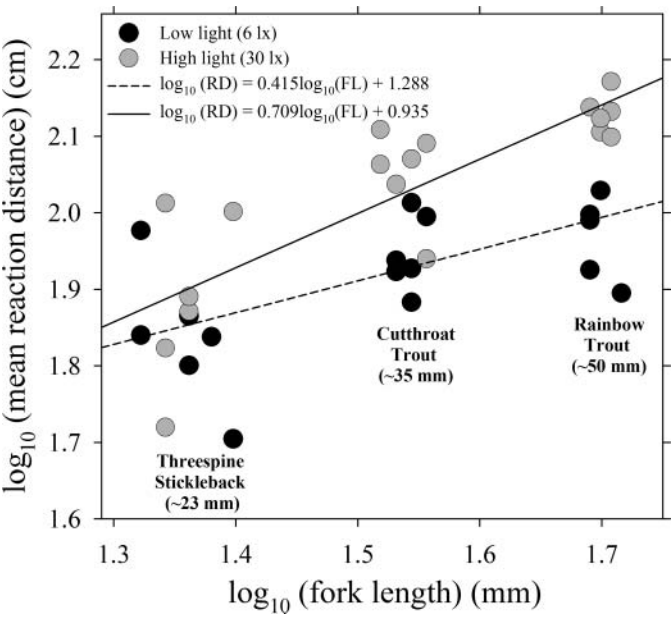


FIGURE 5. Reaction distances (RD) of Chinook Salmon at different combinations of light intensity (lx) and prey size (FL). Points represent the mean RDs observed from replicate trials ($N = 6$) at each combination. Lines represent the best-fit linear models (solid line = high light level; dashed line = low light level). Note that both RD and FL are presented on $\log_{10}$ scales.

TABLE 4. Results from fitting different linear models to test whether reaction distances (RD) of Chinook Salmon varied as a function of prey size and light. Values in bold italics (ΔAIC_c = difference in Akaike's information criterion corrected for small sample size; w_i = Akaike weight) represent the best-fitting models (i.e., ΔAIC_c values ≤ 2). The total number of parameters (k) in each linear model includes the error term (ϵ); l = log-likelihood.

Linear model	k	$-2l$	AIC_c	ΔAIC_c	w_i
$RD \sim \text{Prey Size} + \text{Light} + (\text{Prey Size} \times \text{Light}) + \epsilon$	5	-89.534	-77.534	0.00	0.5396
$RD \sim \text{Prey Size} + \text{Light} + \epsilon$	4	-86.499	-77.208	0.33	0.4584
$RD \sim \text{Prey Size} + \epsilon$	3	-73.087	-66.337	11.20	0.0020
$RD \sim \text{Light} + \epsilon$	3	-58.327	-51.577	25.96	0.0000

Previous studies of piscivorous salmonids have relied on both of the piecewise function types explored in this analysis (i.e., linear and power functions) to describe reaction distance as a function of light (Vogel and Beauchamp 1999; Mazur and Beauchamp 2003). Although beyond the scope of this study, further analyses are warranted to evaluate whether a single functional form and a single SIT value are transferable among species and to quantify resulting errors in estimates of reaction distance from use of an alternative model for different species. However, visual foraging models for piscivores typically approximate search volume by a cylinder with a radius equal to the reaction distance and a length equal to the product of the predator's swimming speed and time spent foraging (Beauchamp et al. 1999). Since estimates of search volume depend on the square of the reaction distance within this formulation, visual foraging models are quite sensitive to errors in estimates of reaction distance. Therefore, when refining the visual foraging approach, generalizations regarding species-specific responses should be explored cautiously.

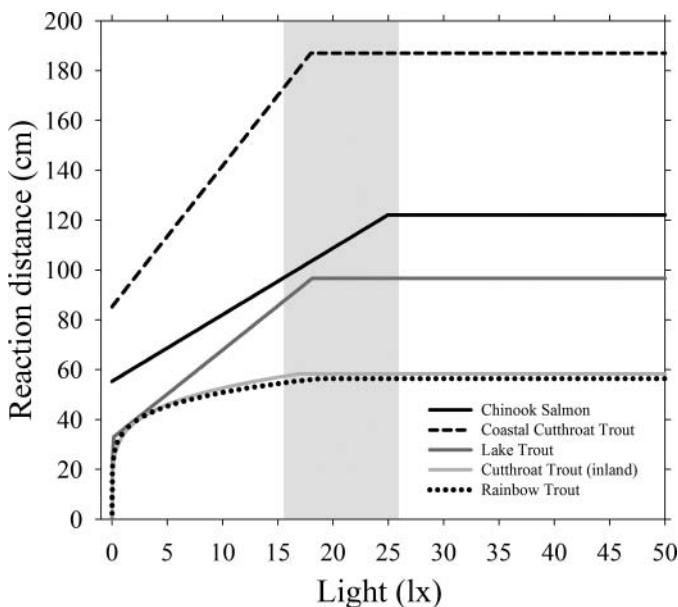


FIGURE 6. Light-dependent reaction distance functions (under clear water conditions) for all piscivorous salmonids evaluated to date. Lines for Chinook Salmon and Coastal Cutthroat Trout are from the present study; data for all other species are from Mazur and Beauchamp (2003). The gray-shaded region brackets the range of estimated saturation intensity threshold values.

The observed threshold responses are important because they mark points above or below which reaction distance is no longer influenced by changes in the optical environment. As long as piscivores occupy depths with ambient light levels above the SIT (e.g., >24.9 lx for Chinook Salmon and >18.0 lx for Coastal Cutthroat Trout), increasing light offers no additional advantage in terms of prey detection. Similarly, prey detection is not limited by turbidity until it increases beyond a critical level (1.65 NTU). Therefore, under conditions in which predator search is not limited, other components of the predation sequence or prey behavior become more important in mediating the magnitude of piscivory. For example, prey evasion is enhanced under improved visibility (Petersen and Gadomski 1994; Mazur and Beauchamp 2003; Meager et al. 2006), and the evasion distances of prey often exceed the reaction distances of piscivores under these conditions (Howick and O'Brien 1983; Miner and Stein 1996).

Although the functional forms of reaction distance in response to light and turbidity observed in this study and other studies were similar across species, the magnitude of the responses differed considerably. Reaction distances above the SIT averaged approximately 50 cm for Rainbow Trout and Bear Lake-strain Bonneville Cutthroat Trout (Mazur and Beauchamp 2003) and about 100 cm for Lake Trout (Vogel and Beauchamp 1999; Figure 6). In contrast, the reaction distances reported here averaged approximately 120 cm for yearling Chinook Salmon and approximately 185 cm for Coastal Cutthroat Trout. It is important to note that our measurements represent an average distance under optimal visual conditions rather than the maximum possible reaction distance. For example, the Coastal Cutthroat Trout occasionally reacted to prey that were over 300 cm away. These distances are quite impressive when considering that the predators were responding to a single, 50-mm prey fish that blended in well with the gray background (our personal observation). The ability of fish to manipulate their coloration to better mimic their surroundings is common (reviewed by Leclercq et al. 2010) and helps to minimize contrast. For successful prey detection by a piscivore, this apparent contrast must exceed the contrast threshold of the piscivore's visual system (Muntz 1990; De Robertis et al. 2003).

Several factors could explain the differences in reaction distance measurements between our study and the studies by Vogel and Beauchamp (1999) and Mazur and Beauchamp

(2003). First, the differences could reflect dissimilarities in the experimental arena. For example, we used a circular tank, whereas a narrower rectangular trough was used in the previous studies. The greater space in the circular tank could have influenced recognition of orientation responses differently. Additionally, reaction distances can vary considerably depending on where the prey is first detected within the visual field of the predator (e.g., transverse versus lateral plane; Dunbrack and Dill 1984). Although not evaluated in the present study, our larger, circular arena provided a greater breadth of angles from which the predators could orient toward the prey, likely allowing the predators to utilize a greater fraction of their visual field. Lastly, we tethered a single prey inside a clear tube in such a manner that enabled restricted swimming around a central pivot point, thereby offering a range of aspects to the predators. The previous studies confined one to three free-swimming prey to a predominately side aspect within an aquarium. Despite these methodological differences, the variability in reaction distances reported in the three studies may reflect real differences in visual capabilities among salmonid taxa.

The differences in reaction distance between the two species examined in this study could reflect pressures associated with fulfilling different life history strategies (i.e., semelparity versus iteroparity) or functional role as a piscivore. For example, anadromous Coastal Cutthroat Trout are iteroparous and remain nearshore in the marine environment (Trotter 1989); Chinook Salmon are semelparous and pelagic, and they require rapid marine growth to optimize the tradeoff between mortality risk and body size for reproduction (Quinn 2005). Given these added pressures and operating under the assumption that reaction distance is largely a behavioral response, we expected that Chinook Salmon would respond to prey at greater distances (both above and below the SIT) than the Coastal Cutthroat Trout. Since eye diameter or area of the retina is correlated with body size and is inherently linked to the theoretical range at which objects can be visually resolved (Aksnes and Giske 1993), physiological mechanisms could also be contributing to differences in reaction distance between the two species. Chinook Salmon reach adult body sizes that are much larger than both the yearling and adult Coastal Cutthroat Trout examined here. It is possible that the reaction distance of Chinook Salmon increases with length or age, reflecting a visual or behavioral ontogeny of prey recognition. For logistical reasons, we could not measure responses from larger Chinook Salmon, and thus we could not evaluate this hypothesis directly. Therefore, our measurements in this study were confined to representing the capabilities of Chinook Salmon at a size where early foraging success is critical for overall marine survival. Conversely, we were able to evaluate a large range of sizes for the Coastal Cutthroat Trout, including smaller juveniles with lengths overlapping those of the yearling Chinook Salmon. Because predator FL did not influence reaction distance for either species over the range of lengths examined, our results show that (1) differences in reaction distance were driven primarily by a species effect when comparing similarly

sized Chinook Salmon and Coastal Cutthroat Trout; and (2) reaction distances of Coastal Cutthroat Trout did not exhibit an ontogenetic response over a broad range of lengths at which this species can be highly piscivorous (Duffy and Beauchamp 2008).

Observed differences in reaction distance were probably not attributable to interspecific differences in spectral sensitivity because both the absorbance spectra for rods and cones and the spectral sensitivity curves are quite similar among salmonid species (Parkyn and Hawryshyn 2000; Novales-Flamarique 2005). Results of experiments by Henderson and Northcote (1985) provide empirical support for this view. They measured the effect of different colors of light (blue, green, yellow, and red) on the reaction distances of Coastal Cutthroat Trout and Dolly Varden *Salvelinus malma* when presented with artificial prey. The effect of color on reaction distance was minimal (within the error calculated for our measurements) for both species across the dominant range of wavelengths emitted by our lamps. The qualitative pattern in the effect of color on reaction distance was the same for both species, but Coastal Cutthroat Trout still exhibited higher reaction distances across all light conditions. Therefore, Henderson and Northcote (1985) concluded that differences in reaction distance were from an effect unrelated to the spectral sensitivity of the species.

Salmonid predators in the present study could perceive all wavelengths of light emitted by the fluorescent lamps (Parkyn and Hawryshyn 2000; Novales-Flamarique 2005), so it is important to consider which fraction of the light spectrum was measured by our light sensors and how this relates to the estimates of reaction distance. We measured light levels in lux using a photometric sensor because (1) this method allowed for direct comparison with previous experiments that used salmonids, (2) the measure incorporated wavelengths of light to which pelagic coastal marine piscivores are generally most sensitive (500–600 nm; Horodysky et al. 2010) with a higher relative response (Biggs 1986), and (3) the lux is a more intuitive unit of measure based on vertebrate vision for evaluating differences among treatments and the extent to which visual conditions change in pelagic environments. The spectral responsivity curve of the photometric sensor matches the International Commission on Illumination (i.e., Commission Internationale de l'Éclairage [CIE]) photopic curve developed for the human eye (Biggs 1986). However, fish are generally more sensitive to shorter, bluer wavelengths than are humans (Horodysky et al. 2010). Although our lux measurements discounted wavelengths in the blue region to a greater extent than likely perceived by the predators, these wavelengths were adequately captured by our corresponding measurements of photosynthetically active radiation (Biggs 1986). Given that the colors of light emitted from our fluorescent lamps should not have had a meaningful influence on reaction distance (Henderson and Northcote 1985), our measurements captured the true light environment experienced by the predators. When considering the purpose and utility of visual foraging models (Beauchamp et al. 1999), both types of

measurements (visible radiation in lux and photosynthetically active radiation in $\mu\text{E}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$) are sufficient for quantitatively describing behavioral responses (e.g., reaction distance) as a function of light.

Twilight is a period of high activity for piscivores (Helfman 1986), and peaks in piscivory under such conditions in lakes have been reported (Beauchamp 1990; Beauchamp et al. 1992; Malmquist et al. 1992; Kahilainen et al. 2009). During dusk, light declines rapidly with depth over short time scales to levels that are well below the SIT values reported here (Henderson and Northcote 1985). Schools of prey often disperse during this time, thereby increasing predator encounter rates with individual prey (Beauchamp et al. 1999; Mazur and Beauchamp 2006). Therefore, the accurate measurement of reaction distances to individual prey under degraded visual conditions is of particular importance. These measurements could be a better indicator of predatory performance than measurements that are obtained at light levels above the SIT. An important question that remains is how the reaction distance changes with aggregations of prey. However, the scale at which these types of experiments can be performed efficiently in the laboratory is likely insufficient for unbiased measurement. By linking telemetry with multibeam acoustics, Dunlop et al. (2010) observed Lake Trout exhibiting rapid bursts of speed toward schools of Ciscoes *Coregonus artedii* (proxy for attack) at proximities of 2.4–6.4 m during daylight in Lake Opeongo, Ontario. Their study represents an important first step toward the in situ measurement of such responses.

Turbidity strongly influences prey detection by piscivores. An increase in turbidity from 0.4 to 7.2 NTU reduced the reaction distance of yearling Chinook Salmon by about 70%. For comparison, the reaction distance of adult Lake Trout diminished by approximately 50% over the same turbidity range (Vogel and Beauchamp 1999). Direct observation by De Robertis et al. (2003) suggested that reaction distances of yearling Sablefish *Anoplopoma fimbria* feeding on Chum Salmon *O. keta* declined by approximately 85% as turbidity increased from 0 to 10 NTU. These results show that the ability of piscivores to effectively detect fish prey declines rapidly with moderate increases in turbidity even under well-illuminated conditions. The range of turbidities used in this experiment reflected a wide range of oligotrophic to eutrophic habitats for salmonids, excluding the higher turbidities experienced in glacial systems and some estuarine waters (Gregory 1994; Gregory and Levings 1998; Vogel and Beauchamp 1999). In productive coastal marine habitats during dense phytoplankton blooms (Lovvorn et al. 2001) or in river plumes (Emmett et al. 2004) with turbidities that may exceed those examined here, planktivory should be a more effective foraging strategy than piscivory for juvenile Chinook Salmon.

Prey size influenced measurements of reaction distance in our experiments. For yearling Chinook Salmon, reaction distances declined significantly by about 30–40% depending on light level as prey length declined from 50 to 23 mm. This experiment was

intended only as an initial examination of the effect of declining prey size on the reaction distances of piscivorous salmonids, and we cannot determine whether differences among prey species contributed to this decline. However, our results suggest that the sizes of postlarval prey fish that have achieved full pigmentation can limit the reaction distance, contrary to what was observed over a range of larger prey in a previous study (55–139 mm; Vogel and Beauchamp 1999). Since all of the prey fish in our study were fully pigmented, this decline in reaction distance could reflect reductions in the cross-sectional area of the prey as length decreased (Aksnes and Giske 1993). Body depth measurements from the prey in the three size treatments supported this suggestion: body depth ranged from 4 to 5 mm for the Threespine Sticklebacks, from 5 to 6 mm for the Coastal Cutthroat Trout, and from 9 to 11 mm for the Rainbow Trout. The transition from a contrast-based visual system to an acuity-based visual system for detecting prey (Breck 1993) may depend more on the onset of pigmentation for translucent larval fish than on prey length alone.

Foraging models that incorporate light-dependent reaction distance functions like those developed here provide a useful mechanistic approach for exploring distribution, foraging success, and predation risk in pelagic communities (Scheuerell and Schindler 2003; Jensen et al. 2006; Gjelland et al. 2009; Kahilainen et al. 2009; Hansen et al. 2013). This study increases the capacity of the visual foraging approach by supplying key parameters for application to additional ecologically and economically important species. Knowledge of how the environment influences prey detection by and foraging success of different piscivore species benefits fisheries managers by providing the means for quantitatively gauging (1) which species may have a competitive or predatory advantage under certain conditions or (2) how the performance of any particular species and resulting predator–prey dynamics are likely to change with shifts in the environment. Our results highlight the need to evaluate species- and life-stage-specific responses. These differences have important implications for predicting how the magnitude of piscivory is likely to change throughout a predator's ontogeny and across environmental conditions, communities, and systems.

One of the goals of most fisheries management agencies is to provide a suite of recreational fishing opportunities for the public while also conserving native species. Different species have been successfully introduced, both legally and illegally, into many salmonid-dominated western lakes in an attempt to meet these competing demands, and some introductions have greatly diminished the ecological integrity of the native fish communities. For example, introductions of Lake Trout into Yellowstone Lake, Flathead Lake, Bear Lake, and Lake Chelan have threatened or diminished the native salmonid populations (Martinez et al. 2009; Ellis et al. 2011; Schoen et al. 2012), requiring that fisheries biologists develop creative management solutions for either reintroducing or conserving the native species (Al-Chokhachy et al. 2009; Hansen et al. 2010). Additionally, the illegal or unintended introduction of nonnative warmwater

piscivores into western systems is an increasing problem (Johnson et al. 2009). Therefore, studies that evaluate the light- and turbidity-dependent prey detection capabilities of key pelagic warmwater piscivores (e.g., Walleye *Sander vitreus* and Striped Bass *Morone saxatilis*) are greatly needed. Armed with this type of information for key species, we will be better equipped for evaluating the potential success and food web implications of native species reintroductions and the ecological impacts of current or looming illegal introductions of nonnative piscivores.

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